

DATA QUALITY ASSESSMENT REPORT

GRO Shield — Independent Governance Assessment

Subject: UGRO Capital Limited (Synthetic Dataset)

Assessment Date: 06 July 2026

Prepared by: GRO Shield Governance Framework

Overall DQ Score: 97.4 / 100

1. Executive Summary

This report presents the findings of an independent data quality assessment conducted on a synthetic MSME loan dataset modeled on the data categories publicly described by UGRO Capital Limited for its GRO Score 3.0 credit scoring model. The assessment evaluated 15,000 loan application records across six internationally recognised data quality dimensions: Completeness, Accuracy, Validity, Uniqueness, Consistency, and Timeliness.

The dataset achieved an overall DQ score of 97.4/100. The most significant finding is in the Completeness dimension, where 8.83% of records contain missing values across bank statement and GST data fields — the two external data source categories most critical to GRO Score model inputs. Underwriting decisions made on incomplete bureau, bank statement, or GST data carry elevated credit risk and potential fair lending exposure.

Three priority recommendations are made: (1) implement automated completeness monitoring with a 99% threshold on all model input fields, (2) establish a data freshness policy requiring bureau and GST data pulls within 30 days of application date, and (3) deploy format validation checks on PAN and GSTIN fields at the point of data ingestion rather than post-processing.

2. Scope and Methodology

The assessment was conducted on a 15,000-row synthetic dataset structured to mirror UGRO Capital's publicly disclosed data categories: credit bureau data, bank statement data, and GST data, supplemented by loan application metadata and outcomes. No real customer, credit bureau, or financial data was used at any stage.

A ground-truth methodology was employed: data quality issues were deliberately injected across all six dimensions before validation, with every injection logged. This enabled calculation of suite recall — the percentage of known issues the validation suite actually detected — producing a recall rate of 94.3% across 2,370 injected issues. Validation was performed using Great Expectations v1.x.

3. Findings by Dimension

Completeness	Score: 91.2/100	GE: FLAGGED
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8.83% of records contain missing values, concentrated in bank statement fields (avg_monthly_balance, avg_monthly_credit_turnover) and GST filing data. The most likely root cause is upstream data pipeline failures — PDF bank statement parsing errors and GSTN API sync delays for recently registered businesses. Missing model inputs require fallback scoring logic; without it, affected applications may be incorrectly declined or approved on partial information.

Accuracy	Score: 97.6/100	GE: FLAGGED
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2.38% of records contain inaccurate values including bureau scores outside the valid 300-900 range, negative loan amounts caused by ETL sign-flip errors, and interest rates with decimal shift errors (e.g. 1800% instead of 18%). While the overall rate is low, inaccurate bureau scores directly corrupt the primary credit signal in GRO Score, making this a high-severity finding despite its relatively low frequency.

Validity	Score: 98.7/100	GE: FLAGGED
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1.29% of records contain values outside defined controlled vocabularies or format specifications. PAN numbers failing the standard 10-character alphanumeric format and state names not matching the approved list indicate free-text contamination from manual data entry or inconsistent source system mappings. Invalid identifiers break downstream KYC and deduplication processes.

Uniqueness	Score: 99.6/100	GE: FLAGGED
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0.40% of applicant IDs are duplicated, indicating primary key collisions likely caused by merge or upsert bugs during data pipeline execution. While the rate is low, duplicate primary keys are a critical data integrity failure — they make it impossible to reliably link records across systems and can result in the same applicant being processed twice.

Consistency	Score: 98.5/100	GE: FLAGGED
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1.50% of records contain cross-field contradictions: approved loan amounts exceeding requested amounts, non-zero approved amounts on rejected applications, and GST registration dates postdating the loan application. These violations indicate insufficient cross-field validation at the point of data entry and create downstream reconciliation failures between the LOS and LMS.

Timeliness	Score: 98.8/100	GE: FLAGGED
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1.20% of records show application dates suggesting bureau data was pulled more than 90 days before the underwriting decision. Stale credit data is a recognised underwriting risk — a borrower's credit profile can change materially over 90 days. RBI Digital Lending Guidelines implicitly require that data used for credit decisions reflects the applicant's current financial position.

4. Overall DQ Scorecard

Dimension	Issues	Rows Affected	Violation Rate	Score	GE Status
Completeness	1350	1324	8.83%	91.2/100	FLAGGED
Accuracy	360	357	2.38%	97.6/100	FLAGGED
Validity	195	193	1.29%	98.7/100	FLAGGED
Uniqueness	60	60	0.4%	99.6/100	FLAGGED
Consistency	225	225	1.5%	98.5/100	FLAGGED
Timeliness	180	180	1.2%	98.8/100	FLAGGED
OVERALL				97.4/100	

5. Recommendations

P1 — Completeness Monitoring

Implement automated null-value monitoring on all GRO Score model input fields with a 99% completeness threshold. Alert on breach within 1 business day. Establish fallback scoring logic for new-to-credit applicants with no bureau history.

P2 — Data Freshness Policy

Define and enforce a maximum data age policy: bureau data no older than 30 days at point of underwriting decision, GST data no older than 60 days. Automate pull-date validation at ingestion.

P3 — Point-of-Ingestion Validation

Deploy PAN and GSTIN format validation at the data ingestion layer rather than post-processing. Reject malformed identifiers at source to prevent downstream KYC and deduplication failures.

Disclaimer

This report is an independent, self-initiated portfolio project. It is not affiliated with, endorsed by, or commissioned by UGRO Capital Limited. All findings are based on a synthetic dataset generated by the author and do not reflect UGRO Capital's actual data quality posture. No real customer, credit bureau, bank statement, or GST data was used. This report is produced solely to demonstrate data governance assessment methodology for career portfolio purposes.